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(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OFFLINE)

Programme: <B.Tech.(CSE)-AI>, Semester: <6th >

(May, 2026)

Subject Code: BAI 306	Subject: Digital Image Processing
Time: 3 Hours	Maximum Marks: 60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5×8=20)
a)	Define Image Sampling and Quantization. Why are they important in digital image processing?	(2.5)
b)	Explain the concept of 4-connectivity and 8-connectivity between pixels with suitable examples.	(2.5)
c)	State the Convolution Theorem and explain its significance in frequency domain filtering.	(2.5)
d)	List and briefly describe any three noise models used in image restoration.	(2.5)
e)	Differentiate between RGB and HSI color models. Which one is more suitable for image processing and why?	(2.5)
f)	Define coding redundancy in image compression. How does it differ from spatial redundancy?	(2.5)
g)	Explain boundary descriptors and regional descriptors with one example each.	(2.5)
h)	What are patterns and pattern classes in object recognition? Give suitable examples.	(2.5)
	UNIT I	
Q2	a) Explain histogram equalization technique for image enhancement with a suitable example. How does it improve the visual quality of an image? b) Describe the process of smoothing using averaging filters. Compare the effects of 3×3 and 5×5 averaging filters on image quality.	(10)
Q3	a) Explain the fundamental steps involved in digital image processing with a neat block diagram. Discuss the role of each component. b) What are the different types of distance measures between pixels? Derive the Euclidean distance formula and compare it with City-block distance.	(10)

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UNIT II

Q4	a) Derive the 2D Fourier Transform and explain its properties. How is it used for image enhancement in the frequency domain? b) Explain the design and implementation of ideal low-pass and high-pass filters in the frequency domain. Discuss their advantages and limitations.	(10)
Q5	a) Describe the image degradation and restoration model. Explain how Wiener filtering is used for image restoration. b) Compare and contrast the following filtering techniques for noise reduction: Geometric Mean Filter, Arithmetic Mean Filter, and Median Filter. When would you use each?	(10)

UNIT III

Q6	a) Explain the fundamentals of color image processing. Discuss various color models (RGB, CMY, HSI) and their applications in image processing. b) Describe edge detection and boundary detection techniques in image segmentation. Explain the Canny edge detection algorithm with suitable examples.	(10)
Q7	a) Explain the fundamentals of image compression. Differentiate between lossy and lossless compression techniques with suitable examples. b) Describe spatial and temporal redundancy in image compression. How are they exploited in various compression models?	(10)

UNIT IV

Q8	a) Explain the concept of morphological operations in image processing. Describe erosion and dilation operations with suitable examples. b) Discuss the use of Principal Component Analysis (PCA) for image representation and description. How does it help in dimensionality reduction?	(10)
Q9	a) Explain decision-theoretic methods in pattern recognition. Describe the Bayes classifier and its application in object recognition. b) Compare structural methods and decision-theoretic methods for object recognition. What are the advantages and limitations of each approach?	(10)

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